Unit 2 Overview Guide

Check Your Understanding — Answer Key

For open questions, a correct response gives the same values with the reasoning in the student's own words. Justifications must use attributes of the relationship, not just the answer.

#	**Answer**
1	Missing $x = 5$; missing $y = 36$ $(k = 4.5)$
2	$y = 0.75x$; $k = 0.75$ — every cup costs \$0.75, the cost per 1 cup
3	A $(y/x = 4.5$ in every row; B and D have an initial value, C is nonlinear)
4	Line through $(0, 0)$, $(2, 5)$, $(4, 10)$, $(6, 15)$ $(y = 2.5x)$
5	D
6	6, 7.5, 9, 12
7	Line through $(0, 20)$, $(4, 15)$, $(8, 10)$ — starts at 20, falls 1.25 per ride
8	Nonproportional; Proportional; Proportional; Nonproportional
9	B
10	D
11	1, 3, 5, 7.5 $(k = 0.25)$; a 30-pound weight stretches it 7.5 inches
12	A $(k = 3.5; 14 \cdot 3.5 = 49)$
13	9 pounds $(k = 11.25/5 = 2.25; 20.25 \div 2.25 = 9)$
14	Nonproportional — the rate of change is constant (+3 per +1 in x) but y/x is not constant (3.5, 3.25, ≈ 3.17), so the line would not pass through the origin
15	D
16	Nonproportional — y/x differs (4.25, 3.5, ≈ 3.36); the rate is a constant \$3 per mile with a \$2.50 starting charge, so the graph would not pass through the origin
17	C $(k = 1.75; 12 \cdot 1.75 = 21)$
18	(a) $y = 2.5x$ (b) the graph is a straight line that passes through the origin, and y/x is the same (2.5) at every marked point
19	A
20	Track A: $y = 7x$; Track B: $y = 4.5x + 10$. At 6 races: A = \$42, B = \$37 — the second track costs less. (Below 4 races the first track costs less, e.g. 2 races: \$14 vs. \$19.)
21	C
22	Nonproportional; Proportional; Nonproportional; Proportional
23	Not correct — a straight line is proportional only if it passes through the origin. Sample: $y = 2x + 3$ is a straight line, but $(0, 3)$ is on the line, and y/x is not constant.
24	$y = 35x + 60$
25	$y = 40x + 50$

#	**Answer**
26	B
27	$y = 2.5x + 4$
28	C
29	rate $= (48 - 78)/(7 - 2) = -6$ gallons per minute; $b = 78 - (-6)(2) = 90$; $y = -6x + 90$
30	B $(m = (22.75 - 10.25)/(8 - 3) = 2.50;\ b = 10.25 - 2.50(3) = 2.75)$
31	(a) $y = 8x + 12$ (b) $y = 6x + 15$ (c) $y = -4x + 25$ (d) $y = 7x + 2$
32	$y = 2.75x;\ \ b = 0$
33	D
34	rate $= (48 - 76)/(3 - 1) = -14;\ b = 76 + 14 = 90;\ \ y = -14x + 90$
35	Rate -7.5: the level drops 7.5 inches each week. Starting value 55: the level at week 0. $55 - 7.5x = 25$ gives $x = 4$ — week 4.
36	A
37	The equation fits: $7(2) + 5 = 19$ and $7(5) + 5 = 40$ — the student is correct; defend $y = 7x + 5$
38	Tank P: $y = 4x + 30$; Tank Q: $y = 7x + 18$. Greater rate: Q (7 gallons per minute fills faster). Greater starting value: P (30 gallons already in the tank at $x = 0$).
39	B
40	Not a function — the input 3 has two outputs (8 and 1)
41	Function — every input has exactly one output; two inputs sharing an output does not break the rule
42	Not a function — the input 5 maps to two outputs (4 and 7)
43	Not a function — $x = 2$ pairs with both 3 and -1
44	C (only $(4, 7)$ uses a new input)
45	Could add: any point whose x is not 1, 4, or 6, e.g. $(7, 5)$. Could not add: any point reusing one of those inputs, e.g. $(4, 9)$. The difference is whether the x-value repeats.
46	D ($x = 3$ maps to both 2 and 11)
47	Mapping: $2 \to 4$, $3 \to 4$, $5 \to 9$. Function — each input has exactly one arrow leaving it.
48	A (each student has exactly one date of birth; B, C, D pair one input with more than one possible output)
49	Not a function — $(-1, 2)$ and $(-1, -2)$ show the input -1 with two outputs
50	Not correct — every input has exactly one output (12), so it is a function; repeated outputs are allowed. Ordered pairs: $\{(3, 12), (5, 12), (7, 12), (9, 12)\}$ — no x repeats.
51	(a) $(3, 5)$ (b) $2(3) - 1 = 5$ ✓ and $-(3) + 8 = 5$ ✓
52	B $(2 + 3 = 5$ and $3(2) - 1 = 5)$
53	(a) $(5, 60)$ (b) 5 is the number of visits at which both gyms cost the same; 60 is that shared cost, \$60

#	**Answer**
54	Both lines drawn; intersection at $(2, 3)$
55	C
56	Not correct — first equation: $-4 + 6 = 2$ ✓; second equation: $0.5(4) + 1 = 3 \neq 2$. The pair satisfies only one equation, so it is not the solution.
57	D
58	Row $y = 2.5x$: 5, 10, 15. Row $y = 1.5x + 4$: 7, 10, 13. Intersection $(4, 10)$ — at $x = 4$ both equations give the same y, so the graphs meet there.
59	Answers vary. Sample: two gyms' total cost in dollars for x months; 3 is the number of months at which both cost the same, and 27 is that shared cost, \$27. (Any situation with two linear quantities equal at $x = 3$, $y = 27$.)
60	Not correct — second equation: $-5 + 6 = 1$ ✓; first equation: $3(5) - 2 = 13 \neq 1$. The point is on only one line; the solution is the point of intersection, $(2, 4)$.